

HTML Forms & Tables

Lab 4 — Reformatted Notes with Extra Attributes, Controls & Validation Covered

Section	Covers
1	Document structure: DOCTYPE, html, head, meta, title, body
2	Grouping form content: fieldset & legend
3	The form tag and its attributes
4	Building tables: table, tr, td, th, caption, thead/tbody
5	Table layout attributes: colspan, rowspan, cellpadding, cellspacing
6	Connecting labels to controls: label, for, id, name
7	The input tag and its types
8	Other form controls often missed: select
9	textarea and button
10	Validation attributes — required, pattern, minlength/maxlength, step
11	Where this fits in the course workflow (HTML -> CSS -> Bootstrap -> Tailwind)
12	Revision quiz

1. Document Structure Tags

Tag	Purpose
<!DOCTYPE html>	Tells the browser this is an HTML5 document. Written before <html>, not a real tag itself.
<html>	The root element — every other tag is written inside it.
<head>	Holds page information (metadata, title, links to CSS) that isn't shown in the body.
<meta>	Provides metadata — e.g. character encoding and responsive viewport settings.
<title>	Sets the text shown on the browser tab.
<body>	Holds all visible page content — the form and table live here.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Feedback Form</title>
</head>
<body>
  <!-- form and table go here -->
</body>
</html>
```

Recap / Extra: lang="en" on <html> declares the page language for screen readers and translation tools — a small but frequently-asked viva detail.

2. Grouping Form Content — fieldset & legend

<fieldset>: Draws a bordered box around a group of related controls — useful when several fields belong to one section of a form.

<legend>: Provides a caption/title for a <fieldset>. It must be placed as the first child inside the fieldset.

```
<fieldset>
  <legend align="center">STUDENT DETAILS</legend>
  <!-- form controls go here -->
</fieldset>
```

Attribute	Belongs to	Meaning
disabled	<fieldset>	Disables every control inside the fieldset at once
name	<fieldset>	Names the group; rarely needed
align (legacy)	<legend>	Old presentation attribute — use CSS text-align instead today

3. The <form> Tag and Its Attributes

Purpose: Defines a form used to collect user input; the controls inside it are submitted together as one unit.

Attribute	Meaning
action	URL that the submitted form data is sent to
method	How data is sent — GET (visible in URL) or POST (sent in the request body)
style	Applies inline CSS directly to the form

```
<form action="/submit-feedback" method="post">  
  ...  
</form>
```

Example: This directly connects back to Lab 1 — submitting this form with method="post" is exactly the POST request you demonstrated with GET vs POST earlier in the course.

4. Building Tables

Tag	Purpose
<table>	Creates the table; used here to neatly align labels and form controls
<tr>	One table row — normally holds <td> or <th> cells
<td>	A normal data cell — can hold text, labels, or entire form controls
<th> (new)	A header cell — bold and centered by default, marks a column/row heading
<caption> (new)	A title for the whole table, placed as the first child of <table>
<thead>/<tbody> (new)	Groups header row(s) separately from body rows — improves structure and styling hooks

```
<table border="1" cellpadding="8" cellspacing="4" align="center">
  <caption>Student Feedback Form</caption>
  <thead>
    <tr><th>Field</th><th>Value</th></tr>
  </thead>
  <tbody>
    <tr><td>Name</td><td>Yug</td></tr>
  </tbody>
</table>
```

Recap / Extra: The original notes only used <td> for every cell, including header-style labels. Using <th> for header cells (and <thead>/<tbody> to group them) is considered better practice and is a common follow-up viva question.

5. Table Layout Attributes

Attribute	Meaning
colspan	Makes one cell span multiple columns — e.g. colspan="2"
rowspan	Makes one cell span multiple rows — e.g. rowspan="2"
cellpadding	Space INSIDE a cell, between its content and its border
cellspacing	Space BETWEEN two adjacent cells
border	Legacy way to add a visible table border
align / valign (legacy)	Old horizontal/vertical alignment attributes — CSS is preferred today

```
<td colspan="2">Email*</td>  
<td rowspan="2">Common</td>
```

Recap / Extra: Memory trick: CELLPADDING = padding is inside a cell. CELLSPACING = gap between cells. COLSPAN = columns. ROWSPAN = rows. This exact pairing is a very common viva question.

6. Connecting Labels to Controls

<label>: Displays descriptive text for a form control. Clicking the label focuses/selects its associated control, and it improves accessibility for screen readers.

```
<label for="email">Email</label>
<input type="email" id="email" name="email">
```

Attribute	Meaning
for (on <label>)	Must match the id of the control it describes
id (on the control)	A unique identifier used by the label, CSS, and JavaScript
name (on the control)	Identifies the field when data is submitted; radio buttons sharing a name form one group

Example: In `<label for="male">Male</label>` paired with `<input id="male" name="gender">`, for and id must match exactly (it's a text match, not a type check) — a common source of a broken form when the two are misspelled differently.

7. The <input> Tag and Its Types

Purpose: Creates a form input control. The type attribute decides what kind of control is displayed and how the browser validates it.

```
<input type="text" id="firstName" name="firstName" placeholder="Enter your first name" required>
```

7.1 Types covered in the original notes

Type	Behaviour
text	Single-line text such as a name
email	Email input with built-in format validation
password	Text is masked while typing
radio	Selects one option from a group sharing the same name
checkbox	Independent on/off selection; several can be checked at once
number	Numeric values only; min/max can limit the range
tel	A field intended for a telephone number
range	Displays a slider; min/max define its range
file	Lets the user choose a file from their device

7.2 Additional types worth knowing (not in the original notes)

Type	Behaviour
date	Shows a native date picker; value is returned as YYYY-MM-DD
color	Shows a native colour-picker swatch
url	Validates that the entered text looks like a web address
search	A text field styled for search, often adding a clear ('x') button
hidden	Not shown to the user, but still submitted with the form

```
<label for="dob">Date of Birth</label>
<input type="date" id="dob" name="dob">

<label for="favColor">Favourite Colour</label>
<input type="color" id="favColor" name="favColor">
```

7.3 for, id, and name — must know

```
<label for="male">Male</label>
<input type="radio" id="male" name="gender" checked>
```

Here for="male" and id="male" match, connecting the label to this exact input. name="gender" groups every radio button that shares it, so selecting one automatically deselects the others in the same group.

8. Other Form Controls Often Missed

These weren't in the original Lab 4 notes but are extremely common in real forms and are a frequent gap in student submissions.

8.1 <select> and <option> — dropdown list

```
<label for="course">Course</label>
<select id="course" name="course">
  <option value="">-- Choose a course --</option>
  <option value="bca">BCA</option>
  <option value="bsc-it">B.Sc IT</option>
</select>
```

9. textarea and button

9.1 <textarea>

Purpose: Creates a multi-line text input — useful for longer information such as a bio, address, or feedback comments.

```
<textarea id="bio" rows="5" cols="30" placeholder="Write your feedback"></textarea>
```

Attribute	Meaning
rows / cols	Approximate visible number of text lines / character columns
placeholder	Temporary hint shown inside the empty textarea
required	Makes the textarea required before successful form submission

9.2 <button>

Purpose: Creates a clickable button. Inside a form, the common types are submit, reset, and button.

```
<button type="submit" style="background-color: teal; color: white;">SUBMIT</button>
<button type="reset">CLEAR</button>
```

Type	Behaviour
submit	Submits the form
reset	Clears all form controls back to their default values
button	A plain button with no built-in form behaviour — used with JavaScript

10. Validation Attributes

Attribute	Meaning
required	Blocks submission until the field has a value
checked	Preselects a radio button or checkbox
placeholder	Temporary hint text shown inside an empty control
min / max	Minimum/maximum allowed value for number, range, and date-type inputs
pattern	A regular expression the value must match — e.g. a specific pin-code format
minlength / maxlength	Minimum/maximum number of characters allowed in a text field
step	The increment allowed for number/range inputs — e.g. step="5"

```
<label for="pin">PIN Code</label>  
<input type="text" id="pin" name="pin" pattern="[0-9]{6}" maxlength="6" required>
```

Recap / Extra: min/max were already in your notes for number/range inputs. pattern, minlength/maxlength, and step extend the same idea to text fields and finer numeric control — worth knowing since they come up often in real registration forms.

11. Where This Fits in the Course Workflow

Legacy HTML attribute (this lab)	Modern CSS replacement (next lab)	Bootstrap/Tailwind equivalent (later labs)
align	text-align / margin	Bootstrap: text-center Tailwind: text-center
border (on <table>)	border property	Bootstrap: table-bordered Tailwind: border
cellpadding / cellspacing	padding / border-spacing	Bootstrap: table classes Tailwind: p-2, border-spacing-2
style="background-color: teal"	a CSS class in a stylesheet	Bootstrap: btn-primary Tailwind: bg-teal-600

Example: Keep this Lab 4 form and table as-is for now — in the CSS lab you'll strip out these inline/legacy attributes and recreate the same look with a stylesheet, which is exactly what this table maps out.

12. Revision Quiz

1. What is `<fieldset>` and what does `<legend>` do inside it?
 2. Is `<legend>` mandatory inside a `fieldset`? Where must it appear if used?
 3. What does the `for` attribute on `<label>` connect to, and how must the values match?
 4. What is the difference between `id` and `name` on a form control?
 5. If the `type` attribute is omitted on `<input>`, what type is used by default?
 6. Explain `cellpadding` vs `cellspacing` with the inside/between memory trick.
 7. Explain `colspan` vs `rowspan` with an example each.
 8. What does `required` do, and how is it different from `pattern`?
 9. What is the difference between `<select>` and `<datalist>` for offering choices to a user?
 10. Why would you use `<th>` and `<thead>` instead of just `<td>` for a table's header row?
 11. What does `method="post"` on a `<form>` do differently from the default `GET`?
 12. Give an example of an input type introduced in this document that wasn't in the original notes.
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